2 **(a)** The pressure p of an ideal gas of density ρ is related to the mean square speed $<c^2>$ of its molecules by the expression

$$p = \frac{1}{3}\rho < c^2 > .$$

Show that the average kinetic energy of a molecule of an ideal gas is proportional to the thermodynamic temperature T.

[3]

(b) A scuba tank for a diver has a volume of $9.4 \times 10^3\ \text{cm}^3$ and when the tank is filled, the air has a pressure of $2.32 \times 10^7\ \text{Pa}$ at a temperature of 24 °C. The diver is swimming in water of density $1.03 \times 10^3\ \text{kg m}^{-3}$ and temperature 24 °C at a depth of 15 m. When the diver breathes in, the pressure of the air delivered from the tank to the diver is always equal to the pressure of the surrounding water.

Atmospheric pressure is $1.01 \times 10^5\ \text{Pa}$. Air may be considered as an ideal gas.

Calculate, for the depth of 15 m,

(i) the total pressure on the diver,

total pressure = ... Pa [2]

(ii) the volume of air available at this pressure from the tank.

volume = ... cm^3 [2]

(c) The supply of air in **(b)** is sufficient for the diver to remain at a depth of 15 m for 45 minutes.

Assuming that the diver always breathes at the same rate regardless of the pressure, determine how long the air in the tank will last for the diver at a depth of 35 m and a water temperature of 19 °C.

time = ... min [3]

3 Fig. 3.1 shows a pair of identical loudspeakers A and B placed 2.0 m apart and emitting